Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/18/2026
PI Name:	Beheshti, Afshin Ph.D.		
Project Title:	Predicting Predisposed Space Radiation from Gravity and Hyperbaric Alterations Determined from Mini-Pigs Studies		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Cancer**:Risk of Radiation Carcinogenesis (3) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (4) **CNS**:Risk of Acute (In-flight) and Late Central Nervous System Effects from Radiation Exposure		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	beheshti@pitt.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-308-2540
Organization Name:	University of Pittsburgh		
PI Address 1:	4200 Fifth Ave		
PI Address 2:			
PI Web Page:			
City:	Pittsburgh	**State:**	PA
Zip Code:	15260	**Congressional District:**	12
Comments:	PI has moved from NASA Ames Research Center to the University of Pittsburgh, where he serves as Director of the Center for Space Biomedicine. Former email: afshin.beheshti@nasa.gov		
Project Type:	Ground	**Solicitation / Funding Source:**	2020-2021 HERO 80JSC020N0001-HHCSR, Omnibus2. Human Health Countermeasures and Space Radiation Topics Appendix C; OMNIBUS2-Appendix D
Start Date:	10/21/2021	**End Date:**	12/31/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Holly, Chapapas Ph.D. (KBR/NASA Johnson Space Center) Costes, Sylvain Ph.D. (Blue Marble Space Institute of Science) Cerqueira, Bianca Ph.D. (Wyle Laboratories, Inc.) Lorenz, Giovanni D.O. (United States Department of the Air Force) Sherman, Paul M.D. (USAF School of Aerospace Medicine)		
Grant/Contract No.:	80NSSC25K7598		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	It is essential that for long-term space travel to fully understand the biological changes that will arise from all components from the space environment. Three main components of the space environment is hypobaria, space radiation, and gravity alterations that systemically impact the body. Both space radiation and gravity alterations have been shown to provide major impact on health during space travel and also provide additive effects to the biology when combined. Hypobaria has been less studied in the context of spaceflight and we believe will have an equal contribution to health risks. Further we believe that hypobaria and/or hyperoxia will sensitize tissues and accelerate space radiation health risks. We hypothesize that gravity alterations and hypobaric changes will provide an environment that will increase space radiation cardiovascular (CVD), Central Nervous System (CNS), and carcinogenesis health risks which can be determined from translational studies from mini-pigs. Specifically, we will study the systemic impact of hypobaria and gravity alterations driven by circulating microRNAs (miRNAs) that are in the majority of the bodily fluids. We have recently identified that a miRNA spaceflight signature can drive universal health risks and is related with space radiation biological alterations. We believe that components of this miRNA signature is associated with specific CVD, CNS, and carcinogenesis health risks. In addition, these circulating miRNAs can be triggered from earlier events to sensitize tissues for increased damage from space radiation, similar to what we have shown with the case of circulating miRNAs related to cancer. Thus, alteration in hypobaria and/or hyperoxia is hypothesized to trigger the miRNA signature associated with spaceflight and provide chronic increase in damage to allow for increased health risks due to space radiation. To provide proof for our hypothesis we will utilize existing data and tissues from hypobaria studies in minipigs which have generated transcriptomic data, imaging data, and biospecimen data including blood, CSF, and tissue from Co-Is, Drs. Cerqueira, Sherman, and Lorenz. This model can also be used to approximate EVA conditions (multi-hour duration at 0.39 atm and 100% O2). The minipig data/analysis will be compared to existing rodent data available on NASA's GeneLab platform for cross-species conserved spaceflight-associated factors between mini-pigs and rodent models and extrapolate to space radiation health risks. Minipigs have many advantages for research over the rodent model which includes the following: Greater tolerance of non-steroidal anti-inflammatory drugs; similar digestive system to humans, which allows for easier comparisons for human drug testing and also microbiome studies; similarities between the human and minipig skin allows for an ideal dermal toxicity model; the gastrointestinal system, digestion and metabolism, and the renal and immune systems are similar to humans, which will allow for easy translational studies for many countermeasure experiments; and importantly, the close similarities to humans for the cardiovascular physiology which include ventricular performance, electrophysiology and coronary artery distribution allows for the most accurate biological testing for cardiovascular risk associated with the space environment. We will achieve the goals of this proposal through the following specific aims: 1) To determine cross-species conserved spaceflight-associated factors between minipigs and rodent models and extrapolate to space radiation health risks. 2) Establish functional significance of circulating miRNAs and neural structural and cardiovascular changes with associated signaling pathways key for hypobaria and synergistic response to space radiation and determining acceptable health risk levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work is innovative by utilizing a systems biology approach to determine whether circulating miRNA signatures influence health risks in the space environment related to hypobaria and microgravity. If the aims are accomplished, this proposal will yield a detailed understanding of the miRNA mediated response to the space environment as well as identify potential novel countermeasures.
Task Progress:	During this reporting period, we completed the foundational data integration and analytical framework necessary to achieve both Specific Aims. Major activities included: • Acquisition and preprocessing of existing datasets from mini-pig hypobaria/hyperoxia studies and murine space radiation/microgravity experiments, including internal archived datasets and curated NASA GeneLab datasets for comparative analyses. • Generation of cross-species differential expression maps using RNA-seq and miRNA-seq data from serum, CSF, and brain tissues. • Systems-level bioinformatics analyses including principal component analysis (PCA), differential gene expression, Gene Set Enrichment Analysis (GSEA), and Ingenuity Pathway Analysis (IPA), to identify pathways and signatures perturbed by spaceflight-relevant conditions. • Integration of neuroimaging, transcriptomic, and biomarker profiles from the mini-pig model to establish functional links with circulating miRNAs and CNS injury phenotypes.
Bibliography Type:	Description: (Last Updated: 07/13/2026)